

Online Retail Data Cleaning and Exploratory Data Analysis Report

Project Overview

This project involved cleaning, analyzing, and visualizing the Online Retail dataset to uncover sales trends, customer purchasing behavior, product performance, and revenue patterns. The dataset initially contained 541,909 transaction records and required extensive cleaning before analysis could be performed.

Data Cleaning Challenges Encountered

Several data quality issues were identified during the cleaning process:

- Missing values were found in the Description and CustomerID columns.
- Duplicate transaction records were identified and removed.
- Cancelled transactions were detected through invoice numbers beginning with the letter "C".
- Records with non-positive unit prices were identified and removed.
- Column names, text fields, and date formats were standardized to ensure consistency.

These cleaning actions improved data quality and ensured that subsequent analyses were based on valid sales transactions.

Key EDA Findings

The exploratory analysis revealed several important business patterns:

- PAPER CRAFT, LITTLE BIRDIE was the most purchased product based on quantity sold.
- DOTCOM POSTAGE generated the highest revenue among all products.
- The United Kingdom was the highest revenue-generating country, contributing more than £9 million in sales.
- Monthly revenue peaked in November 2011, generating approximately £1.5 million.
- Customer purchasing behavior was highly concentrated among a relatively small group of frequent buyers.

Top Insights

1. The United Kingdom is the retailer's primary market and contributes the majority of total revenue.
2. Product popularity does not always translate into the highest revenue, as pricing significantly influences financial performance.
3. Revenue demonstrates strong seasonal patterns, with sales peaking during the holiday shopping period.
4. A small group of highly active customers contributes disproportionately to transaction volume.
5. High-value products play a critical role in overall business performance and should be monitored alongside sales volume metrics.

Conclusion

The cleaned dataset provided valuable insights into customer behavior, product performance, sales trends, and revenue generation. The findings demonstrate how data analysis can support informed business decisions related to inventory management, customer retention, pricing strategies, and market expansion opportunities.